

NIKKO, Japan

WMO: 476900

Lat: 36.7397N

Lon: 139.5003E

Elev: 1294

StdP: 86.72

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-10.8	-9.7	-18.1	0.9	-6.6	-16.7	1.0	-6.7	14.3	-3.0	13.0	-2.9	3.9	260	0.531

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	6.3	25.1	20.3	24.0	20.0	22.8	19.4	21.5	23.8	20.7	22.8	20.0	21.9	3.0	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
20.7	18.0	22.9	20.1	17.3	22.1	19.4	16.6	21.3	69.5	23.9	66.7	23.0	64.0	21.9	23.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)		
(4) 11.9	9.9	8.4		-12.7	27.5	1.1	1.0	-13.5	28.2	-14.1	28.8	-14.8	29.4	-15.6	30.2	
(5)				-13.6	22.6	1.0	0.6	-14.4	23.0	-15.0	23.3	-15.6	23.7	-16.3	24.1	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	7.3	-4.2	-3.7	-0.4	5.1	10.4	14.0	18.4	19.2	15.4	9.7	4.3	-1.1
	DBStd	8.66	2.93	3.43	3.65	3.67	2.80	2.67	2.44	2.22	3.13	3.15	3.31	3.39
	HDD10.0	1888	441	383	322	151	28	2	0	0	2	43	173	343
	HDD18.3	4103	699	617	581	397	246	132	29	17	97	267	421	601
	CDD10.0	912	0	0	0	4	41	122	262	284	162	35	2	0
	CDD18.3	86	0	0	0	0	0	2	33	43	7	0	0	0
	CDH23.3	167	0	0	0	0	1	3	69	86	8	0	0	0
	CDH26.7	4	0	0	0	0	0	0	2	3	0	0	0	0
(14) Wind	WSAvg	3.2	4.4	4.2	4.0	3.5	2.9	2.3	2.2	2.2	2.3	2.6	3.5	4.2
(15) Precipitation	PrecAvg	1832	47	49	92	137	164	196	244	262	307	184	98	53
	PrecMax	2334	159	132	187	257	390	574	435	565	666	523	322	175
	PrecMin	1135	7	10	21	30	63	67	108	82	32	12	2	6
	PrecStd	277	36	30	39	69	72	97	91	134	160	112	70	37
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	6.9	9.3	12.6	18.5	21.7	23.5	26.6	26.9	24.4	20.0	14.8	10.7
		MCWB	3.7	5.0	6.1	10.4	14.0	17.8	20.1	20.6	19.0	15.3	9.4	7.3
	2%	DB	4.2	6.3	9.9	15.2	19.4	21.4	25.1	25.4	22.6	17.9	12.7	7.7
		MCWB	1.1	2.6	4.9	9.0	12.5	16.9	20.4	20.6	18.7	14.8	8.9	4.7
	5%	DB	2.4	3.8	7.6	13.3	17.6	20.0	23.9	24.2	21.2	16.2	11.1	6.0
		MCWB	-0.5	0.7	3.9	8.2	12.1	16.3	20.0	20.4	18.1	13.5	8.2	2.9
	10%	DB	0.8	1.8	5.7	11.5	16.0	18.6	22.7	23.0	19.8	14.7	9.7	4.3
		MCWB	-1.9	-0.8	2.5	6.9	11.4	15.7	19.4	20.0	17.6	12.4	7.3	1.7
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	4.4	6.2	8.2	13.3	16.4	19.6	22.4	22.4	20.6	17.1	11.8	8.7
		MCDB	6.0	8.6	10.3	14.9	18.8	21.5	24.8	25.0	22.6	18.4	12.8	10.0
	2%	WB	1.9	3.4	6.2	11.0	14.7	18.2	21.5	21.7	19.6	15.6	10.3	5.5
		MCDB	3.7	5.6	8.5	13.8	17.0	20.1	23.8	24.0	21.3	17.1	11.5	7.0
	5%	WB	-0.2	1.2	4.6	9.4	13.5	17.3	20.7	21.0	18.8	14.3	8.9	3.5
		MCDB	1.8	3.2	6.9	11.7	16.0	19.1	22.9	23.1	20.3	15.6	10.3	5.3
	10%	WB	-1.7	-0.4	3.0	8.0	12.3	16.3	19.9	20.3	18.1	12.9	7.6	1.9
		MCDB	0.5	1.5	5.3	10.3	14.8	17.9	21.7	22.3	19.5	14.2	9.2	3.8
(35) Mean Daily Temperature Range	5% DB	MDBR	6.7	7.4	7.9	8.8	8.7	6.7	6.2	6.3	6.2	7.0	7.4	6.7
		MCDBR	8.5	9.8	10.4	11.9	11.5	8.9	8.3	8.4	7.6	8.4	8.9	8.7
	5% WB	MCWBR	6.8	7.7	7.7	8.0	7.5	5.7	5.1	5.2	4.8	6.0	6.9	6.8
		MCWBR	8.1	9.0	9.7	10.2	9.7	7.1	7.5	7.6	6.6	7.0	8.0	8.2
(40) Clear-Sky Solar Irradiance	taub		0.303	0.337	0.394	0.437	0.459	0.489	0.493	0.492	0.423	0.379	0.334	0.307
		taud	2.294	2.182	2.086	2.050	2.035	2.008	2.071	2.068	2.234	2.300	2.382	2.376
	Ebn at Noon		881	887	864	846	831	803	796	788	827	833	840	854
		Edh at Noon	97	124	151	165	171	175	163	161	129	110	90	85
(44) All-Sky Solar Radiation	RadAvg		2.34	3.08	3.97	4.67	5.14	4.67	4.56	4.59	3.72	3.02	2.37	2.08
	RadStd		0.15	0.22	0.25	0.41	0.52	0.36	0.70	0.50	0.31	0.28	0.19	0.14

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (37 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page